

Yiğit Sarp Avcı

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Education

Boğaziçi University - B.Sc. in Computer Engineering

09/2023 – Present

- Extracurriculars: Executive Board Member & DJ at Radyo Boğaziçi, Regional Organizational Team Member at ESTIEM; Student-Athlete (Sailing, Skiing & Windsurfing).

Kabataş Boys' High School

09/2018 – 06/2023

- Graduated with a GPA of 3.78/4.00; Extracurriculars: Debate, Model United Nations, Entrepreneurship, and Coding Club.

Experience

Software Engineering Summer Intern at Smash Mate

06/2026 – 07/2026

- Completed an intensive 1-month summer internship during the critical pre-launch phase, contributing to extensive backend and frontend repositories.
- **Full-Stack Exposure:** Contributed to a complex microservices-oriented backend (Java, Quarkus) and a mobile frontend application (React Native).
- **Feature Development & Quality:** Developed core backend features, resolved high-priority issues, and authored comprehensive unit/integration tests to ensure zero-defect deployments.
- **Launch Readiness:** Played an active role in agile workflows, code reviews, and system optimizations leading directly to the product's successful official launch.

Work and Travel Program Participant

06/2025 – 09/2025

- Gained international work experience in the USA, improving English communication, teamwork, and adaptability in a multicultural environment.

Educational Mentor at Derece Kampüsü

06/2023 – 09/2025

- Gave mentorship and guidance to students preparing for university entrance exam, helping them with academic and career development.

Technical Projects (@GitHub)

FinTech & Trading Systems — Python, Next.js, FastAPI, PostgreSQL, Alembic, Pytest, WebSockets, Docker

- Developed **AlphaTrigger** (a real-time trading simulator featuring a Z-Score mean reversion strategy and volatility-adjusted position sizing) and the **FinTech Intelligence Platform** (featuring a live limit-order matching engine, portfolio risk analyzer, and AI-powered insights).

Enterprise Dashboards & Fullstack — Next.js 15+, TypeScript, Express, Prisma, TypeORM, Tailwind CSS, SWR

- Developed **Performo**, a real-time administrative dashboard with role-based access control, task management, and optimized database queries, alongside an **Enterprise FAQ System** featuring SWR-powered optimistic UI updates, Express backend, and robust API error boundaries.

Systems & CPU Scheduling — x86-64 Assembly, GNU Toolchain, Linux Syscalls, Process Scheduling, Multithreading (Pthreads), Memory Management

- Implemented **SchedSim**, a CPU scheduling simulator written in x86-64 GNU Assembly using direct Linux syscalls to execute FCFS, SJF, SRTF, Priority, and Round Robin scheduling algorithms.

Interpreters & Language Design — C, Python, Bash, Lexer Design, Functional Paradigms

- Built **Spireslayer** and **Aura**, including a C-based command interpreter with custom backtracking lexer/parser and a Python functional-language interpreter with closures, higher-order functions, and lexical/dynamic scoping.

Algorithms & Matching Engines — Java, Graph Theory, Heaps, Complexity Analysis

- Developed **MatrixNet** (a network graph analysis tool implementing Tarjan's SCC algorithm and articulation point detection) and **GigMatch-Pro** (a candidate matching engine using max-heap data structures for priority-based job allocation).

Mobile Development, Game & WebAssembly Engineering — React Native, Expo, C++, Qt6, WebAssembly, Java, TypeScript, Reanimated, Audio API, CI/CD

- Built **Smash Minesweeper** (a polished mobile app featuring a reducer-driven state engine, BFS flood-fill, and custom audio effects), **Qt 2048** (a C++/Qt6 clone cross-compiled to WebAssembly with undo history), and **Nightpass** (a terminal-based survival card game with a state-machine rule engine).

Skills

- **Tools, AI & Ecosystem:** Linux/Unix, Git/GitHub, AI-Assisted Development, ccache, Makefile, LaTeX, MS Office
- **Languages:** Turkish (native), English (fluent), German (elementary)

Volunteer Experience

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| ▪ TOÇEV: Provided volunteer tutoring and mentoring for children from disadvantaged backgrounds. | 09/2019 – 03/2020 |
| ▪ Boğaziçi University Industrial Engineering Club: Actively contributed to the Volunteering Division. | 09/2023 – 06/2024 |
| ▪ Kabataş Boys’ High School CIP Club: Organized and executed various community-oriented projects. | 09/2018 – 06/2023 |